# BRAINDANCER · CONTROLS

every button, knob, layer, combo and page

firmware v0.9 · M-VAVE FM-1 · norfolkhacker.github.io/braindancer

## NAVIGATION · SIX RULES

### 1 · SELECT WALKS THE PAGES

Every page of the section in one chain, a page a detent, stopping at both ends. No levels, no zoom.

FX: SENDS → SPACE → MASTER → MORE

### 2 · THE BUTTON: IN, OR ON

From another section: the page you were last on there. Already inside: the next page (round).

SEQ › STEPS: SEQ → STEP EXTRAS · GLO › LIVE: GLO → MIX

### 3 · PRESETS LOADS PRESETS

Always: the track's sounds; SEQ the patterns (PHRASES, OCT+ loads); GEN the recent pieces. Never a page, a track or a module.

FX › SPACE: PRESETS ▶ the next sound

### 4 · KNOB 1 PICKS

The step, the slot, the stage, the operator, the kit's sound, the row. The track: always ALGORITHM.

SEQ › STEPS: K1 = all 64 steps · MORE: K1 = the row, K2 its value

### 5 · HOLD = THE LAYER

Held + SELECT: HOME GEN · EDIT SEQ LFO FX their dice · GLO the tempo · ENV SCL ARP SAVE their chain.

SEQ + G5 CLR STEP · SAVE + C5 NAME

### 6 · THE SCREEN SAYS WHERE

The header names the section and the track; the status line what the knobs and the next tap do. Footer: the page's place, n / N.

SEQ · STEPS 1 / 4

## THE PANEL

### KNOBS AND TRANSPORT

| | |
|---|---|
| **MASTER** | Master volume. |
| **SELECT** | **Walks the section's pages**, a page a detent, stopping at the ends (the MENU: its pages). Held with a button: HOME **GEN**; EDIT · SEQ · LFO · FX their **dice** (OCT+ keeps, OCT– back); GLO the tempo; ENV · SCL · ARP · SAVE their chain. |
| **PRESETS** | **Always presets**: the sounds (factory and user) on every page; SEQ: the patterns (PHRASES); GEN: the recent pieces. Nothing in the MENU or with a button held. |
| **ALGORITHM** | Selects the track from any page: T1–T6 (T1–T4 with TRACKS 4). MENU: nothing. A layer held: MAP · KNOBS · STATE. |
| **KNOB 1–4** | The four knobs on screen, in their colours; KNOB 1 picks (step, slot, row). Fast sweeps the whole range, slow fine-tunes. With a layer held: the layer's. A step held: a lock on that step. |
| **OCT– · OCT+** | The keyboard's octave (both together: back to 0); on DRUM the kit stays put: +1/+2/+3 REPEAT 1/2, REPEAT, RATCHET X2; –1 SHORT, –2 GHOST, –3 FLAM. On action rows and dialogs: back / confirm. |
| **PLAY** | Start / stop. GLO + PLAY: restart from the top. |
| **REC** | Arms live recording on the selected track (notes and knob moves, overdub). On SEQ › STEPS stopped: step recording. |
| **HOME** | Tap: HOME (on HOME the next page: TRACKS · VISUAL · GEN). **Hold: the MENU.** |
| **KEYS** | 27 keys play the selected track; with a layer held they become that layer's; on GLO › LIVE they are the tricks. |

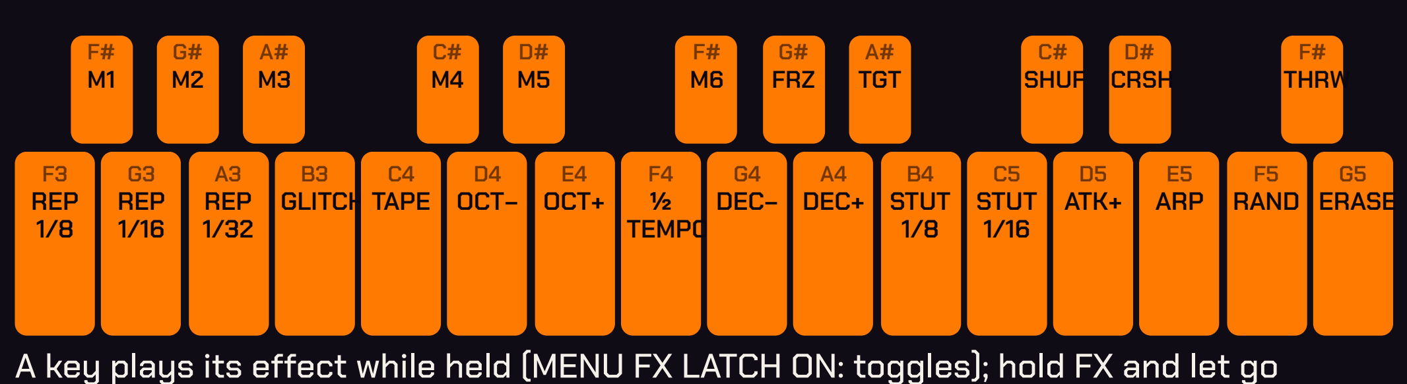
### SECTIONS

| BUTTON | PAGES (A TAP: THE NEXT · SELECT WALKS THEM) | HOLD |
|---|---|---|
| HOME | TRACKS · VISUAL · GEN (ENERGY · FORM · COMPLEX · TAPE; OCT+ GENERATE; PRESETS RECENT) | MENU |
| EDIT | SOUND · TONE · VOICE or FILTER · MORE (FM6: SOUND · OPERATOR · OP ENV · MORE; DRUM-X: KIT · SOUND · SOUND 2 · MORE) | INIT · RECALL · sound mutes |
| SEQ | STEPS · STEP EXTRAS · PATTERN · MORE (TRACK · STEP · MUTATE · SHAPE · PLAYHEADS · AUTOMATION) | SEQ TOOLS |
| FX | SENDS · SPACE · MASTER · MORE (DISTORT · AUTO PAN · DELAY · REVERB · CHORUS · SLICER · SLICE PITCH · PUNCH LANE · MASTER · TAPE) | PUNCH-IN FX |
| GLO | LIVE · MIX · MIDI · MORE (LIVE: the rails · SYSTEM · TEMPO · MIDI · UNDO); at power-on LIVE | MUTE · SOLO · LEVEL · TAP · UNDO |
| ENV | AMP · ENV TO | — |
| LFO | LFO · DEST · MOD MATRIX · MORE (PHASE · STEPS) | MACROS M1–M4 (stay latched) |
| SCL | KEY · CHORD · MORE | ROOT by key |
| ARP | ARP · ARP 2 | — |
| SAVE | SOUND · SONG · PROJECT · MORE (from outside, SOUND comes with SAVE picked) | SONG layer · C5 NAME |
| REC | (transport) | CLEAR · CLICK |

BUTTON from elsewhere: the page last shown; inside: the next page. SELECT walks every page of the section. **MORE**: KNOB 1 picks a row, KNOB 2 sets it.

## LAYERS · HOLD A BUTTON

### FX · PUNCH-IN EFFECTS



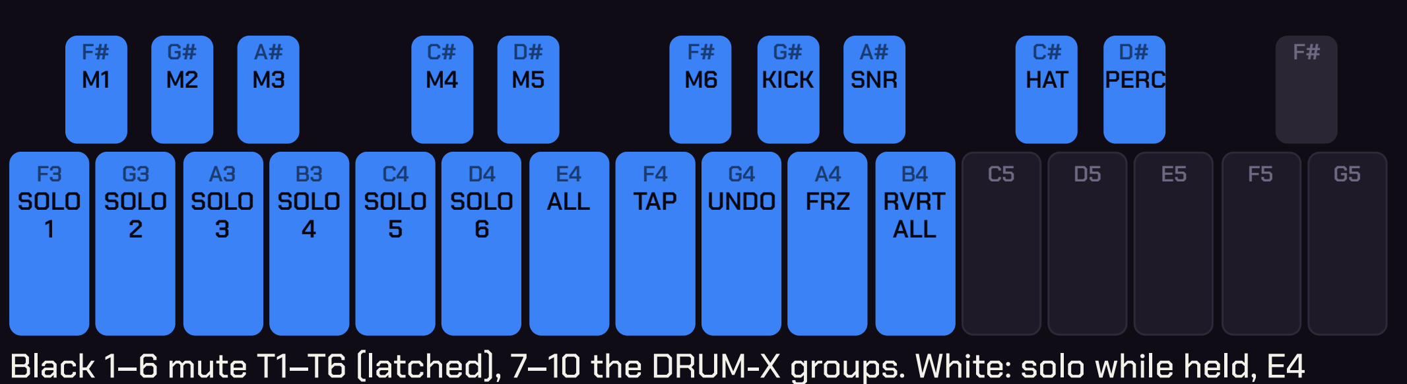
A key plays its effect while held (MENU FX LATCH ON: toggles); hold FX and let go untouched: **LOCKED** open, FX again unlocks; OCT– all off. In on the next 1/16. Black 1–6 mute T1–T6, G#4 FREEZE, A#4 picks which tracks the note effects touch (all · synths · drums), C#5 SHUFFLE; D#5 F#5 sweep CRUSH, THROW dry to wet over 2 bars (back in a beat). With REC armed and playing they record into the section's lane; G5 erases it.

KNOBS · PERFORMANCE MACROS

| | | | |
|---|---|---|---|
| STEP LOOP | INTERVAL | MUTATE | DJ FILTER |

Up to the right; turn any back: all four off at once. GLITCH, TAPE STOP, FREEZE, SHUFFLE held: their own four.

### GLO · MIX



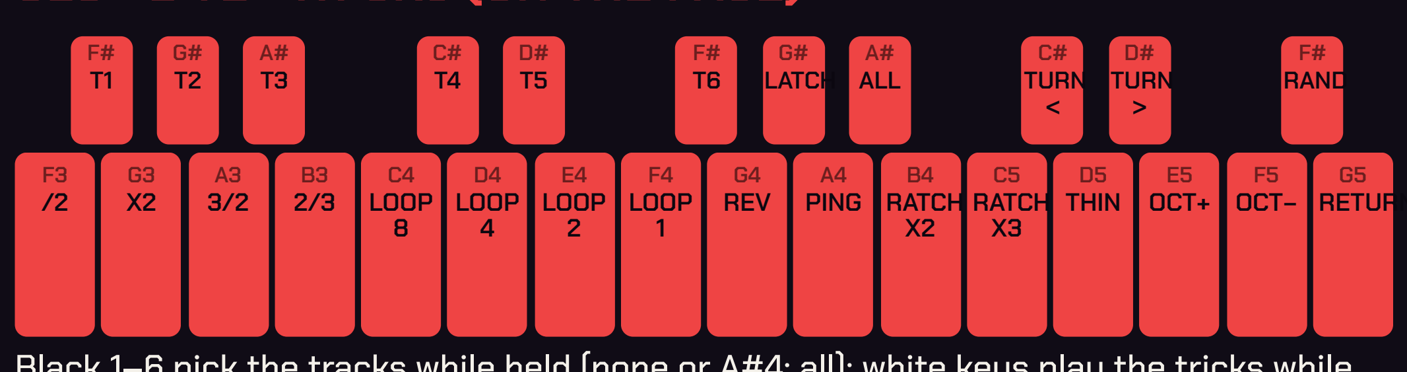
Black 1–6 mute T1–T6 (latched), 7–10 the DRUM-X groups. White: solo while held, E4 unmutes all, F4 tap tempo, G4 **UNDO** the last big change (again: redo), A4 **FREEZE** MUTATE, B4 **REVERT ALL** (every track back to MUTATE's anchor). SELECT: the tempo. PLAY: restart.

KNOBS

| | | | |
|---|---|---|---|
| LEVEL T1 | LEVEL T2 | LEVEL T3 | LEVEL T4 |

T5 or T6 selected: the knobs are T3–T6.

### GLO › LIVE · TRICKS (ON THE PAGE)

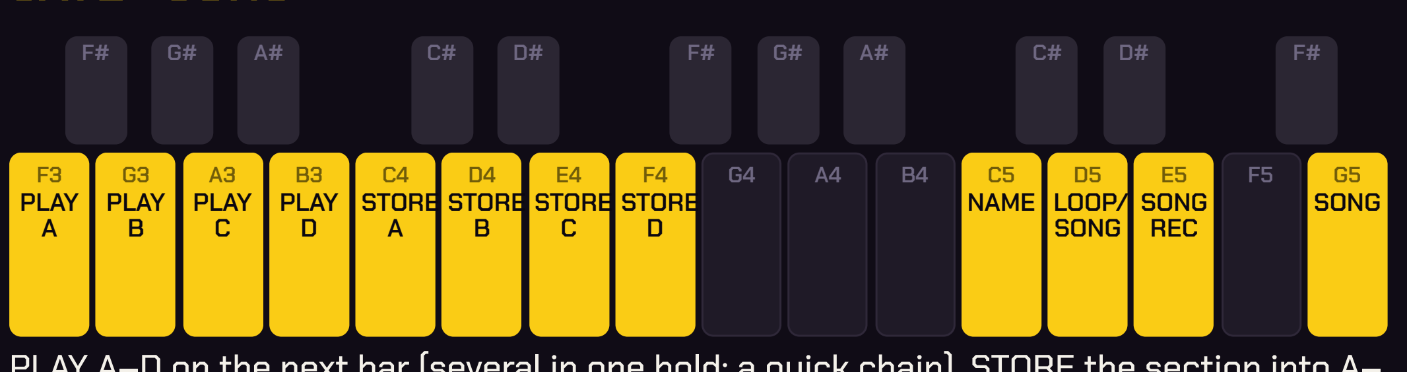


Black 1–6 pick the tracks while held (none or A#4: all); white keys play the tricks while held. G#4 LATCH, C#5 / D#5 TURN, F#5 RANDOM. OCT– tapped RETURN, held BYPASS; OCT+ twice KEEP (into the sequence; GLO + G4 undoes). GLO › MORE › LIVE: the rails (TRACKS, RATE, CUT, DENSITY), latched.

KNOBS

| | | | |
|---|---|---|---|
| RATE | LOOP | THIN | DJ FILTER |

### SAVE · SONG

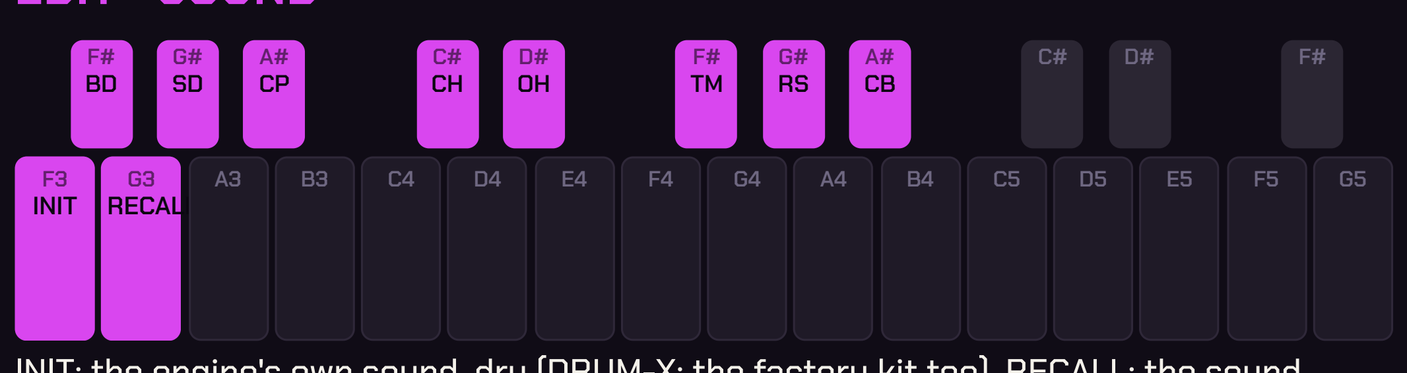


PLAY A–D on the next bar (several in one hold: a quick chain), STORE the section into A–D. OCT+ stores the playing section, OCT– recalls it. C5 renames (SOUND, SONG, PROJECT). On SONG KNOB 4 stores a scene.

KNOBS

| | | | |
|---|---|---|---|
| PROJECT | - | - | - |

### EDIT · SOUND



INIT: the engine's own sound, dry (DRUM-X: the factory kit too). RECALL: the sound stored in the section. On a DRUM track black 1–8 mute each DRUM-X sound.

KNOBS

| | | | |
|---|---|---|---|
| SOUND 1 | SOUND 2 | SOUND 3 | SOUND 4 |

### LFO · MACROS

Knobs M1–M4; they stay latched when let go (LFO again: off). F3 rolls new routes, G3 clears them (only tracks that sound). LFO + SELECT: the track's LFO diced.

KNOBS

| | | | |
|---|---|---|---|
| M1 | M2 | M3 | M4 |

### SCL · ROOT

Any key sets the ROOT. ROOT and SCALE are every melodic track's.

KNOBS

| | | | |
|---|---|---|---|
| ROOT | SCALE | ROTATE | PITCH |

### SEQ · TOOLS

On the SEQ pages: F3 CLEAR · G3 REVERSE · A3 B3 SHIFT · C4 RANDOM · D4 COOK · E4 BEAT · F4 CLR AUTO · G4–C5 the lane's tools (DRUM) · D5 REVERT · G5 CLR STEP · REC: CLEAR ALL. No undo.

KNOBS

| | | | |
|---|---|---|---|
| LENGTH | DIVISION | SWING | GATE |

### REC · RECORD

F3 CLEAR the track (at once; GLO + G4 undoes it), G3 the metronome CLICK.

KNOBS

| | | | |
|---|---|---|---|
| CLICK | - | - | - |

## COMBOS

### BUTTONS TOGETHER

| | |
|---|---|
| HOME + SELECT | GEN, the song generator (HOME's last pages). |
| EDIT · SEQ · LFO · FX + SELECT | Their dice: sound · sequence · LFO · effects (BEFORE / NOW; OCT+ keeps, OCT– back). |
| ENV · SCL · ARP · SAVE + SELECT | Into that section and along its chain. |
| SEQ + G5 | CLR STEP: the picked step and its locks (the AUTOMATION list: deletes the record). |
| SAVE + C5 | NAME: rename the slot, the section, the project. |
| REC + FX EDIT ENV LFO SCL ARP GLO | Clear that part's automation on the selected track (FX: the punch-in lane too). |
| SEQ + REC | CLEAR ALL SEQUENCES? (OCT+ confirms). |
| SAVE + OCT+ / OCT– | Store / recall the playing section. |
| GLO + PLAY | Restart from the top. |
| GLO + SELECT | Tempo (from any page; under CLOCK USB or TRS the clock's). |
| STEP HELD + KNOB | A lock on that step (LOCK · SETTING · VALUE); + SELECT the EDIT page they lock; + EDIT clears its locks. |
| HOME held | MENU: AUDIO · CLICK · CONTROLS · MORE (speaker EQ, tracks, load, BPM lock; click and count-in; hold time, knob speedup, FX latch, calibration; MORE: screen, LEDs, USB audio, CPU cores and clock, ABOUT). |

## EVERY PAGE · KNOB 1 – 4

### ENV

| | | | | |
|---|---|---|---|---|
| AMP | ATTACK | DECAY | SUSTAIN | RELEASE |
| ENV TO | FILTER | PITCH | SHAPE | LOOP |

### LFO

| | | | | |
|---|---|---|---|---|
| LFO | RATE | WAVE | MODE | FADE |
| DEST | PITCH | FILTER | SHAPE | AMP |
| MOD MATRIX | SLOT | SOURCE | DEST | AMOUNT |
| MORE | PHASE · STEPS (LENGTH, STEP RATE, SMOOTH, STEP 1–16): K1 picks, K2 sets | | | |

### SCL · ARP

| | | | | |
|---|---|---|---|---|
| KEY | ROOT | SCALE | SNAP | ALL TRACKS |
| CHORD | CHORD | VOICING | STRUM | - |
| MORE | TRANSPOSE: K1 picks, K2 sets | | | |
| ARP | ARP | RATE | OCTAVES | GATE |
| ARP 2 | SWING | CHANCE | HOLD | ORDER |

### FX

| | | | | |
|---|---|---|---|---|
| SENDS | DISTORT | CHORUS | DELAY | REVERB |
| SPACE | DLY TIME | DLY FDBK | REV TYPE | REV SIZE |
| MASTER | CLIP | PUNCH | DUCK | AGE |
| MORE | DISTORT · AUTO PAN · DELAY · REVERB · CHORUS · SLICER · SLICE PITCH · PUNCH LANE · MASTER · TAPE: K1 picks, K2 sets | | | |

### EDIT

| | | | | |
|---|---|---|---|---|
| SOUND | E1 | E2 | E3 | E4 |
| TONE | E5 | E6 | E7 | E8 |
| VOICE | VOICE | GLIDE | PRIORITY | SPREAD |
| FILTER | FILTER | CUTOFF | RESO | DISTORT |
| FM6 SOUND | ALGORITHM | FEEDBACK | MOD LEVEL | MOD RATIO |
| OPERATOR | OP | RATIO | FINE | LEVEL |
| OP ENV | OP | STAGE | RATE | LEVEL |
| KIT | MORPH | TUNE | TONE | DECAY |
| DRUM SOUND | DRUM | LEVEL | PITCH | DECAY |
| SOUND 2 | DRUM | NOISE | COLOR | WAVE |
| MORE | the engine's other settings in groups: K1 picks, K2 sets | | | |

### SEQ

| | | | | |
|---|---|---|---|---|
| STEPS | STEP | NOTE | ACC/SLD | GATE |
| STEPS · DRUM | STEP | LANE | HIT | ACC |
| STEP EXTRAS | RATCHET | CHANCE | CONDITION | MICRO |
| PATTERN | LENGTH | DIVISION | SWING | GATE |
| MORE | TRACK (SEQ PITCH, CHANCE, HUMAN, REC SNAP, PLAYS) · STEP (TYPE, TYPE 2, EVERY, FOCUS, PITCH, MOVE) · MUTATE (SOUND, PATTERN, AMOUNT, MODE) · SHAPE (STRETCH, REPEAT, NUDGE, ROTATE) · PLAYHEADS (HEAD, RATE, DIRECTION, INTERVAL) · AUTOMATION (PLAY, EVENTS, LOCKS, CLEAR, the records): K1 picks, K2 sets | | | |

### GLO · SAVE · MENU

| | | | | |
|---|---|---|---|---|
| LIVE | RATE | LOOP | THIN | DJ FILTER |
| MIX | LEVEL | PAN | SWING | TUNE |
| MIDI | IN | OUT | CHANNEL | CLOCK |
| MORE | LIVE (TRACKS, RATE, CUT, DENSITY) · SYSTEM (TRACKS, LOAD, CPU) · TEMPO (BPM) · MIDI (KEYBOARD PLAYS SELECTED, CLOCK OUT, NOTES OUT, LEARN, LEARN CLEAR) · UNDO: K1 picks, K2 sets | | | |
| SAVE · SOUND | SLOT | PRESET | NAME | SAVE |
| SAVE · SONG | SECTION | ROW | BARS | SCENE |
| SAVE · PROJECT | PROJECT | LOAD | SAVE | NAME |
| SAVE · MORE | SONG (END, DELETE ROW, CLEAR SONG) · TOOLS (CLEAR SEQ, INIT SOUND, ERASE USER SLOT, NEW PROJECT, CPU TEST) · SOUND (LOAD SLOT, ENGINE, FAVORITE, LIST): K1 picks, K2 sets | | | |
| MENU | AUDIO (SPEAKER EQ TRACKS LOAD BPM LOCK) · CLICK · CONTROLS (HOLD TIME KNOB SPEEDUP FX LATCH CALIBRATION) · MORE | | | |

## GLO › MIDI

### MIDI PER TRACK · THE CLOCK

KNOB 1 **IN**, KNOB 2 **OUT**, KNOB 3 **CHANNEL** (1–16) of the selected track (ALGORITHM picks it; its row lit in the table), KNOB 4 **CLOCK** INT / USB / TRS. A track plays the notes and CCs of its channel while IN is on, and sends its notes on its channel over USB while OUT and NOTES OUT are on. Channel 16's Program Change always picks a section or project. GLO › MORE › MIDI: **KEYBOARD PLAYS SELECTED** (notes on a channel no track listens to play the selected track), **CLOCK OUT** (USB only: the TRS socket is an input), **NOTES OUT** OFF / KEYS / KEYS+SEQ, **LEARN**, LEARN CLEAR. Under CLOCK USB or TRS the tempo is the clock's.

GPL-3.0 · Built on JIANT FM (Juan Jiant) and Felucca (Hügelton Instruments) · song mode from SLOOP